GNANA PRAKASH MYLA

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Professional Summary

Electrical and Computer Engineering graduate with hands-on experience in embedded systems, sensor integration, signal processing, control systems, and lab-based validation. Skilled in developing microcontroller-based prototypes, processing physiological and sensor data, building MATLAB/Python analysis pipelines, and testing hardware/software systems. Interested in embedded firmware, medical devices, neurotechnology, controls, automation, validation, and research engineering roles.

Education

University of Alabama at Birmingham Birmingham, AL
Master of Science in Electrical and Computer Engineering — GPA: 3.66 *2026*

Relevant Coursework: Embedded Systems, Machine Learning, Edge Computing, Computer Vision, Modern Control Theory, Industrial Controls, Digital Systems

Technical Skills

Programming: C, C++, Python, MATLAB, Verilog, SQL

Embedded Systems: ESP32, Arduino, STM32 basics, ARM Cortex-M basics, Heltec LoRa 32 V3, Arduino MKR WAN 1310, I^2C, SPI, UART, I^2S, GPIO, ADC, PWM

Signal Processing: EEG preprocessing, filtering, FFT, PSD, Welch PSD, bandpower extraction, artifact handling, feature extraction, time-series analysis, noise reduction

Machine Learning: scikit-learn, logistic regression, SVM, LDA, gradient boosting, model evaluation, classification metrics

Controls and Automation: PID, LQR, Kalman filtering, MATLAB/Simulink, Stateflow, PLC/HMI basics, OpenPLC, industrial control concepts

Tools: MATLAB, Simulink, Python, MNE-Python, EEGLAB, Git, KiCad basics, Multisim, Wokwi, Arduino IDE, VS Code, Linux/UNIX

Lab and Validation: Sensor testing, data acquisition, oscilloscope debugging, soldering, hardware bring-up, integration testing, system-level troubleshooting, CSV/log analysis

Research Experience

Volunteer Graduate Researcher – ANRY Lab, UAB Jan. 2026 – Present
Neural Switching System, Embedded Control, Hardware Validation *Birmingham, AL*

- Worked on a 16-channel neural stimulation and recording switching system for controlled channel routing.
- Assisted with architecture planning for Arduino Mega-based relay control, TTL triggering, and channel sequencing.
- Designed switching logic to route one selected channel to stimulation while maintaining safe recording paths for other channels.
- Evaluated hardware constraints including relay timing, settling delays, TTL synchronization, amplifier safety, and fault prevention.
- Contributed to system design discussions involving STG stimulator triggering, BNC routing, RJ45 channel grouping, and GUI/code architecture.

Volunteer Researcher – Neural Signal Processing and Modeling Lab, UAB May 2026 – Present
EEG Signal Processing, MATLAB, Python, Data Analysis *Birmingham, AL*

- Supported EEG-based vigilance and neural signal analysis research using MATLAB, Python, and signal-processing workflows.
- Analyzed EEG datasets to study temporal, spatial, and spectral changes related to vigilance and fatigue.

- Developed feature extraction pipelines using theta, alpha, and beta frequency-band behavior for fatigue-state interpretation.
- Compared EEG-derived vigilance features against PERCLOS-based labels to evaluate reliability across subjects and conditions.
- Prepared visualizations including topographic maps, time-series plots, regional comparisons, and subject-level performance summaries.

Project Experience

AlphaSync Bio – EEG-Based Vigilance and Fatigue Monitoring System 2025 – 2026
MATLAB, Python, EEG, Signal Processing, Machine Learning

- Developed an EEG-based vigilance monitoring concept focused on fatigue detection for safety-critical environments.
- Processed EEG datasets using bandpass filtering, notch filtering, artifact-aware analysis, and sliding-window feature extraction.
- Extracted theta, alpha, and beta-band features to study spatial-spectral changes associated with vigilance variation.
- Compared EEG-based indicators with PERCLOS labels to evaluate fatigue-state alignment and prediction reliability.
- Built subject-level and region-level analysis workflows to study temporal evolution, lag behavior, and cross-subject variability.
- Presented AlphaSync Bio as a neurotechnology-focused research/startup concept involving non-invasive fatigue monitoring.

Acoustic-Based Heart Rate Monitoring Using INMP441 and ESP32 2025
ESP32, I^2S, Embedded C/C++, Python, Edge Machine Learning

- Built a non-contact heart-rate monitoring prototype using an INMP441 I^2S microphone and ESP32-based signal acquisition.
- Captured acoustic signal amplitude data and processed time-series patterns for heartbeat-related peak detection.
- Developed embedded logic to classify signal quality and heart-rate condition into no signal, noisy signal, normal, and tachycardia classes.
- Integrated predefined logistic-regression weights for lightweight edge classification on microcontroller hardware.
- Explored LoRa-based communication for remote physiological monitoring use cases.

Neonatal Chest Motion and Skin Temperature Monitoring System 2025
Arduino MKR WAN 1310, AHT10, MPU6050, I^2C, Python, MATLAB

- Developed a sensor-based monitoring system to capture neonatal chest motion, temperature, and humidity signals.
- Integrated MPU6050 IMU and AHT10 temperature/humidity sensor using I^2C communication.
- Collected synchronized timestamped data for rest, light movement, moderate/high movement, waving, and hand-holding conditions.
- Improved data reliability by adding timestamps, validating sampling consistency, and comparing sensor behavior across motion states.
- Analyzed CSV data using Python/MATLAB to generate plots, compare baseline behavior, and evaluate sensor response during movement.

Computer Vision Classification, Detection, Segmentation, and GAN Project 2025
Python, PyTorch/Keras, YOLO, UNet, Vision Transformer

- Implemented computer vision workflows involving image classification, object detection, segmentation, and generative modeling.
- Compared CNN-based and transformer-based model behavior for visual recognition tasks.
- Applied segmentation models such as UNet/SegNet and evaluated model outputs across different image-processing tasks.
- Developed technical understanding of dataset preparation, model training, evaluation, and result interpretation.

Teaching Experience

Teaching Assistant – University of Alabama at Birmingham Jan. 2025 – Apr. 2025
Electrical and Computer Engineering *Birmingham, AL*

- Assisted students with electrical and computer engineering coursework, lab concepts, and technical problem-solving.
- Supported instruction involving engineering fundamentals, assignments, debugging, and academic guidance.
- Communicated technical concepts clearly to students with different levels of engineering preparation.

Additional Experience

Operations Attendant – UAB University Recreation May 2025 – Jul. 2025
Facility Operations and Student Support *Birmingham, AL*

- Supported facility operations, safety checks, customer service, and daily coordination tasks.
- Developed reliability, communication, and team-support skills in a fast-paced university environment.

Presentations and Recognition

AlphaSync Bio – EEG-Based Vigilance Monitoring Research Concept
Presented/developed for UAB NeuroGateways and external innovation opportunities, focused on non-invasive EEG-based fatigue estimation using spatial-spectral signal features and vigilance-state modeling.

Create the Future Design Contest – AlphaSync Bio
Submission qualified under the medical category.